

```

import pandas as pd

df = pd.read_csv("/content/avaliacao_transactions.csv", parse_dates=

# KPIs gerais
total_transacoes = len(df)
total_fraudes = df["is_fraud"].sum()
taxa_fraude = total_fraudes / total_transacoes * 100

# Fraude por canal
analise_canal = (
    df.groupby("channel")
      .agg(
        total_transacoes=("transaction_id", "count"),
        total_fraudes=("is_fraud", "sum"),
        risco_medio=("risk_score", "mean"),
        valor_total=("amount", "sum")
      )
      .reset_index()
)

analise_canal["taxa_fraude"] = (
    analise_canal["total_fraudes"]
    / analise_canal["total_transacoes"]
    * 100
)

print(analise_canal)

```

	channel	total_transacoes	total_fraudes	risco_medio	valor_total
0	app	12021	455	49.822697	2007907.06
1	atm	3007	54	49.502441	538079.57
2	pos	5968	85	50.593396	1088597.67
3	web	9004	157	50.364439	1573259.20

	taxa_fraude
0	3.785043
1	1.795810
2	1.424263
3	1.743669

✓ Analise feita por categoria

```
analise_categoria = (  
    df.groupby("merchant_category")  
        .agg(  
            total_transacoes=("transaction_id", "count"),  
            total_fraudes=("is_fraud", "sum"),  
            ticket_medio=("amount", "mean"),  
            risco_medio=("risk_score", "mean")  
        )  
        .reset_index()  
)  
  
analise_categoria["taxa_fraude"] = (  
    analise_categoria["total_fraudes"]  
    / analise_categoria["total_transacoes"]  
    * 100  
)
```

✓ Analise feita para segmento

```
analise_segmento = (  
    df.groupby("segment")  
        .agg(  
            total_transacoes=("transaction_id", "count"),  
            total_fraudes=("is_fraud", "sum")  
        )  
        .reset_index()  
)  
  
analise_segmento["taxa_fraude"] = (  
    analise_segmento["total_fraudes"]  
    / analise_segmento["total_transacoes"]  
    * 100  
)
```

✓ Analise da pergunta de negocio elaborada pelo grupo

```

analise_combinada = (
    df.groupby(["channel", "merchant_category"])
      .agg(
          total_transacoes=("transaction_id", "count"),
          total_fraudes=("is_fraud", "sum")
      )
      .reset_index()
)

analise_combinada["taxa_fraude"] = (
    analise_combinada["total_fraudes"]
    / analise_combinada["total_transacoes"]
    * 100
)

analise_combinada = analise_combinada.sort_values(
    "taxa_fraude",
    ascending=False
)

```

```

df["mes"] = df["timestamp"].dt.month
df["hora"] = df["timestamp"].dt.hour

analise_mensal = (
    df.groupby("mes")
      .agg(
          total_transacoes=("transaction_id", "count"),
          total_fraudes=("is_fraud", "sum")
      )
      .reset_index()
)

analise_mensal["taxa_fraude"] = (
    analise_mensal["total_fraudes"]
    / analise_mensal["total_transacoes"]
    * 100
)

```

```

import pandas as pd

# =====
# ETAPA 3 – ANÁLISE DOS DADOS
# Disciplina: Big Data
# Alunos: Gabriel Ramos e Thales Vieira
# =====

```

```
# Leitura da base
df = pd.read_csv(
    "avaliacao_transactions.csv",
    parse_dates=["timestamp"]
)

print("\n===== ETAPA 3 - ANÁLISE TECHPAY =====")
print(f"Total de transações: {len(df)}")

# -----
# 1. KPIs GERAIS
# -----

total_fraudes = df["is_fraud"].sum()
taxa_fraude = (total_fraudes / len(df)) * 100

print("\n--- 1. KPIs GERAIS ---")
print(f"Total de transações: {len(df)}")
print(f"Total de fraudes: {total_fraudes}")
print(f"Taxa geral de fraude: {taxa_fraude:.2f}%")

# -----
# 2. FRAUDE POR SEGMENTO
# -----

segmento = df.groupby("segment").agg(
    transacoes=("transaction_id", "count"),
    fraudes=("is_fraud", "sum")
)

segmento["taxa_fraude"] = (
    segmento["fraudes"] / segmento["transacoes"] * 100
)

print("\n--- 2. FRAUDE POR SEGMENTO ---")
print(segmento.round(2))

# -----
# 3. FRAUDE POR CANAL
# -----

canal = df.groupby("channel").agg(
    transacoes=("transaction_id", "count"),
    fraudes=("is_fraud", "sum")
)

canal["taxa_fraude"] = (
    canal["fraudes"] / canal["transacoes"] * 100
)
```

```
,  
  
print("\n--- 3. FRAUDE POR CANAL ---")  
print(canal.round(2))  
  
# -----  
# 4. FRAUDE POR CATEGORIA  
# -----  
  
categoria = df.groupby("merchant_category").agg(  
    transacoes=("transaction_id", "count"),  
    fraudes=("is_fraud", "sum")  
)  
  
categoria["taxa_fraude"] = (  
    categoria["fraudes"] / categoria["transacoes"] * 100  
)  
  
print("\n--- 4. FRAUDE POR CATEGORIA ---")  
print(categoria.round(2))  
  
# -----  
# 5. ANÁLISE TEMPORAL  
# -----  
  
df["hora"] = df["timestamp"].dt.hour  
  
horario = df.groupby("hora").agg(  
    transacoes=("transaction_id", "count"),  
    fraudes=("is_fraud", "sum")  
)  
  
horario["taxa_fraude"] = (  
    horario["fraudes"] / horario["transacoes"] * 100  
)  
  
print("\n--- 5. FRAUDE POR HORA ---")  
print(horario.round(2))  
  
# -----  
# 6. PERGUNTA DE NEGÓCIO  
# Canal + categoria  
# -----  
  
combinacao = df.groupby(  
    ["channel", "merchant_category"]  
) .agg(  
    transacoes=("transaction_id", "count"),  
    fraudes=("is_fraud", "sum")
```

```

)

combinacao["taxa_fraude"] = (
    combinacao["fraudes"] / combinacao["transacoes"] * 100
)

combinacao = combinacao.sort_values(
    "taxa_fraude",
    ascending=False
)

print("\n--- 6. CANAL + CATEGORIA ---")
print(combinacao.round(2))

print("\n===== FIM DA ANÁLISE =====")

```

===== ETAPA 3 - ANÁLISE TECHPAY =====

Total de transações: 30000

--- 1. KPIs GERAIS ---

Total de transações: 30000

Total de fraudes: 751

Taxa geral de fraude: 2.50%

--- 2. FRAUDE POR SEGMENTO ---

	transacoes	fraudes	taxa_fraude
segment			
High-Risk	2915	282	9.67
Premium	16440	157	0.95
Standard	10645	312	2.93

--- 3. FRAUDE POR CANAL ---

	transacoes	fraudes	taxa_fraude
channel			
app	12021	455	3.79
atm	3007	54	1.80
pos	5968	85	1.42
web	9004	157	1.74

--- 4. FRAUDE POR CATEGORIA ---

	transacoes	fraudes	taxa_fraude
merchant_category			
alimentacao	5976	138	2.31
eletronico	4440	87	1.96
saude	2951	73	2.47
servicos	4576	90	1.97
varejo	9047	203	2.24
viagem	3010	160	5.32

--- 5. FRAUDE POR HORA ---

	transacoes	fraudes	taxa_fraude
--	------------	---------	-------------

hora			
0	1288	40	3.11
1	1233	55	4.46
2	1259	39	3.10
3	1234	41	3.32
4	1277	43	3.37
5	1262	29	2.30
6	1192	32	2.68
7	1284	39	3.04
8	1231	27	2.19
9	1248	22	1.76
10	1291	29	2.25
11	1268	23	1.81
12	1238	26	2.10
13	1246	29	2.33
14	1249	28	2.24
15	1322	31	2.34
16	1171	33	2.82
17	1241	15	1.21
18	1169	17	1.45
19	1280	26	2.03
20	1194	35	2.93
21	1250	39	3.12
22	1231	27	2.19